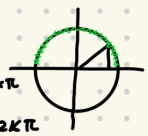
ESERCIZI FOGLIO 3

② $E = \{x \in \mathbb{R} : \sin(18x) > 0\} \cap [0, \pi]$

D: $\sin(18x) > 0$

$\sin y > 0$ $\quad y = 18x$



$0 < y < \pi$

$2k\pi < y < \pi + 2k\pi$

$2k\pi < 18x < \pi + 2k\pi$

$\frac{2k\pi}{18} < x < \frac{\pi + 2k\pi}{18}$

$9k\pi < x < \frac{\pi + 2k\pi}{18}$

$E = \left[\bigcup_{k \in \mathbb{Z}} \left(\frac{2k\pi}{18}, \frac{\pi + 2k\pi}{18} \right) \right] \cap [0, \pi]$

$= (0, \pi) \cup \ldots \cup \left(\frac{2 \cdot 8\pi}{18}, \frac{17}{18}\pi \right)$

$E = \bigcup_{k=0}^{8} \left(\frac{2k\pi}{18}, \frac{\pi + 2k\pi}{18} \right)$

I. E è un intervallo → Falsa perché abbiamo un'unione di intervalli

II. E è limitato → Vero $[0, \pi]$

III. sup(E) è π → Falso è $\frac{17}{18}\pi$

IV. inf(E) è 0 → Vero è 0

③ $a_n = \frac{10n+1}{6n+1}$

$\lim_{n \to \infty} \frac{4n^3+3}{5n+1} = \infty$

$\frac{n^3(4 + \frac{3}{n^3})}{n(5 + \frac{1}{n})} = \frac{4n^2}{5} = \infty$

I. $\lim_{n \to \infty} a_n = 0$ Falsa $\quad \frac{\infty}{\infty}$

$= \lim_{n \to \infty} \frac{n(10 + \frac{1}{n})}{n(6 + \frac{1}{n})} = \frac{10}{6} = \frac{5}{3}$

II. $\lim_{n \to \infty} (a_{n+1} - a_n) = \frac{5}{3}$ Falso

$= \left(\lim_{n \to \infty} a_{n+1} \right) - \left(\lim_{n \to \infty} a_n \right) = \frac{10}{6} - \frac{10}{6} = 0$

III. $\lim_{n \to \infty} (a_{n+1} - a_n)\, n = 0 \cdot \infty$

$a_{n+1} - a_n = \frac{10(n+1)+1}{6(n+1)+1} - \frac{10n+1}{6n+1}$

$= \frac{10n+11}{6n+7} - \frac{10n+1}{6n+1}$

$= \frac{(10n+11)(6n+1) - (10n+1)(6n+7)}{(6n+7)(6n+1)}$

$= \frac{(60n^2 + 10n + 66n + 11) - (60n^2 + 70n + 6n + 7)}{(6n+7)(6n+1)}$

$= \frac{(76n + 11) - (76n + 7)}{(6n+7)(6n+1)}$

$= \frac{4}{12n^2 + 6n + 42n + 7}$

$= \frac{4}{12n^2 + 48n + 7}$

$\lim_{n \to \infty} \left(\frac{4}{12n^2 + 48n + 7} \right) n$

$\lim_{n \to \infty} \frac{n(4)}{n^2(12 + \frac{48}{n} + \frac{7}{n^2})} = \frac{4}{12n} = \frac{1}{3n} = \frac{1}{\infty} = 0$

④ $a_n = \frac{An^2+1}{Bn^2+1}$, $A > 0$, $B > 0$

se $A=6$ $B=13$ $a_n \nearrow$

$a_{n+1} - a_n > 0 \quad \forall n$

$a_{n+1} \geq a_n$

$$\frac{A(n+1)^2+1}{B(n+1)^2+1} - \frac{An^2+1}{Bn^2+1} = \frac{(A(n^2+1)+1)(Bn^2+1) - (An^2+1)(B(n^2+1)+1)}{(B(n^2+1)+1)(Bn^2+1)} =$$

$(B(n^2+1)+1)(Bn^2+1) > 0$ quindi possiamo ignorarlo

$$= 2(A-B)n + (A-B) \geq 0 \quad \forall n \in \mathbb{N}$$

se $n=0 \rightarrow A-B \geq 0$

$A \geq B$

$2\underbrace{(A-B)}_{\geq 0} n + \underbrace{(A-B)}_{\geq 0} \geq 0$

$A > B \Rightarrow a_n \nearrow$

$A = B \Rightarrow a_n$ COSTANTE

$A < B \Rightarrow a_n \searrow$

I. $\lim_{n\to\infty} a_n = +\infty$ Falso

$\lim_{n\to\infty} \frac{n^2(A+\frac{1}{n^2})}{n^2(B+\frac{1}{n^2})} = \frac{A}{B}$

II. $A=10$ $B=7$ $a_n \nearrow$ Vero

III. $A=6$ $B=13$ $a_n \searrow$ Vero

IV. $A=11$ $B=8$ $\nexists$ MAX(E) $E = \{a_n : n \in \mathbb{N}\}$ Falso

$a_n \nearrow$

$\lim_{n\to\infty} a_n = \sup\{a_n\} \stackrel{?}{=} \max(E)$ QUANDO $\frac{11}{8} \in E$

$\lim_{n\to\infty} = \frac{11}{8}$

$\frac{11}{8} = \frac{11n^2+1}{8n^2+1}$

$11(8n^2+1) = 8(11n^2+1)$

$88n^2 + 11 = 88n^2 + 8$

$11 = 8$ impossibile quindi: $\nexists \max(E)$

$b_n \to +\infty$ $a_n \to +\infty$

$a_n \ll b_n$ $\Rightarrow$ $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$

$a_n \smallsetminus b_n$

ESEMPIO

$n^3 \gg n^2$ $\lim_{n\to\infty} \frac{n^2}{n^3} = 0$

$n^\alpha \gg n^\beta \quad \alpha > \beta$

$A > 1$, $\gamma > 0$, $\alpha > 0$, $\beta > \alpha$, $B > 0$, $C > B$

$(\log_A n)^\gamma \ll n^\alpha \ll n^\beta \ll B^n \ll C^n \ll n! \ll n^n \ll 2^n \ll 3^n$

ESEMPI

$$\lim_{n\to\infty} \frac{3^n + n!}{2^n + n^2 + n^n} = \frac{\infty}{\infty}$$

$$\lim_{n\to\infty} \frac{n!\left(\frac{3^n}{n!} + 1\right)}{n^n\left(\frac{2^n}{n^n} + \frac{n^2}{n^n} + 1\right)} = \frac{n!}{n^n} = 0$$

LIMITI NOTEVOLI

$a_n \to 0$

I. $e^{a_n} - 1 \to 0$

II. $\log(1 + a_n) \to 0$

III. $\sin(a_n) \to 0$

IV. $\operatorname{tg}(a_n) \to 0$

V. $\arcsin(a_n) \to 0$

VI. $\operatorname{arctg}(a_n) \to 0$

VII. $1 - \cos(a_n) \to 0$

$\frac{i}{a_n} \to 1$ (i esimo punto tra)

$\frac{①}{a_n} = \frac{e^{a_n} - 1}{a_n}$

$\frac{⑦}{a_n} \to 0 \quad \frac{7}{(a_n)^2} = \frac{1}{2}$

ESEMPIO

1. $\lim_{n\to\infty} \frac{\sin\left(\frac{2}{n}\right)}{\frac{3}{n}} \to " \frac{\sin\left(\frac{2}{n}\right)}{\frac{2}{2}\cdot\frac{3}{n}} \to " \frac{\sin\left(\frac{2}{n}\right)}{\frac{3}{2}\cdot\frac{2}{n}} \to \frac{2}{3}$ ($\frac{2}{n} \to 0$)

$a_n \to 0$

$\frac{\sin(a_n)}{a_n} \to 1$

II. $\lim_{n\to\infty} \frac{\left[\operatorname{tg}\left(\frac{1}{n}\right)\right]^2}{e^{3/n^2} - 1}$

$a_n \to 0$

$\lim_{n\to\infty} \frac{\operatorname{tg} a_n}{a_n} = 1$

$\lim_{n\to\infty} \frac{e^{a_n} - 1}{a_n} = 1$

$$\lim_{n\to\infty} \left[\frac{\operatorname{tg}\left(\frac{2}{n}\right)\cdot\frac{2}{n}}{\frac{2}{n}}\right]^2 \cdot \frac{1}{e^{3/n^2} - 1}$$

$$= " \underbrace{\left[\frac{\operatorname{tg}\left(\frac{2}{n}\right)}{\frac{2}{n}}\right]^2}_{\to 1} \cdot \frac{\frac{4}{n^2}\cdot\frac{3}{3}}{e^{3/n^2} - 1}$$

$$= " \underbrace{\left[\frac{\operatorname{tg}\left(\frac{2}{n}\right)}{\frac{2}{n}}\right]^2}_{\to 1} \cdot \frac{4}{3} \cdot \underbrace{\left[\frac{\frac{3}{n^2}}{e^{3/n^2} - 1}\right]}_{\to 1}$$

$= 4/3$

ESERCIZIO

$$\lim_{n\to\infty} \frac{10\cdot n! + 2\cdot 3^n}{n^n + 4} = 0$$

$$= " \frac{10n! + 6^n}{n^n + 4}$$

$$= \frac{n!\left(10 + \frac{6^n}{n!}\right)}{n^n\left(1 + \frac{4}{n^n}\right)} = \frac{10n!}{n^n} = 0$$

$$\lim_{n\to\infty} \left(1 + \frac{2}{n^3}\right)^8 = 1$$

$$" \left(1 + \frac{A}{b_n}\right)^{b_n} = e^A, \quad b_n \to +\infty$$

$$\lim_{n\to\infty} \left[\underbrace{\left(1 + \frac{2}{n^3}\right)^{n^3}}_{e^2}\right]^{\left(\frac{1}{n}\right)\to 0}$$

$$= (e^2)^0 = 1$$

ESERCIZIO

$$f(x) = \sqrt{x^2 - x - 30}$$

E = INSIEME DI DEF DI $f(x)$ $\rightarrow$ $E = \{x \in \mathbb{R} : x^2 - x - 30 \geq 0\}$

D:

$$x^2 - x - 30 \geq 0$$

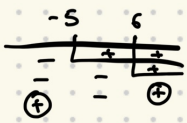
$$\Delta = (-1)^2 - 4(1)(-30) = 121$$

$$x_{1,2} = \frac{1 \pm 11}{2} = 6, -5$$

$$(x-6)(x+5) \geq 0$$

$x - 6 \geq 0 \rightarrow x \geq 6$

$x + 5 \geq 0 \rightarrow x \geq -5$



$x \leq -5 \vee x \geq 6$

$E = \{x \in \mathbb{R} : x \leq -5, x \geq 6\}$

$E = (-\infty, -5] \cup [6, +\infty)$

I. Insieme E è un intervallo? Falso

II. Insieme E è limitato? Falso

III. $F = E \cap [0, +\infty) = [6, +\infty)$ $\exists \max(F)$? Falso

IV. $G = E \cap (-6, 7] = (-6, -5] \cup [6, 7]$ $\exists \max(G)$? Vero 7